Reference Event 8: The Collapse of the Francis Scott Key Bridge, March 2024

Characterization Against the Transportation Sector Design Basis

DBA-TRAN-RE-8 — Reference Event Instrument, DBA-TRAN Series

Lineage: Issued under the Transportation Sector Framework v0.10; DBA-MA Parent Framework; conferred on the proposal of Ways and Linehaul (DBA-TRAN-WY-FC2)

Version 0.2 — Working Draft

October 2026

Timothy E. Roxey

Eclectic Technologies

Held by Project 11 — DBA Reference Events · released for review and discussion

SUGGESTED CITATION

T. Roxey, "DBA-TRAN-RE-8 — Key Bridge Collapse," Version 0.2, in CIKR Design Basis Examples, Edition 1 (November 2026), Eclectic Technologies, November 2026. https://doi.org/10.5281/zenodo.23863986

This version is a member of CIKR Design Basis Examples, Edition 1 (November 2026) (https://doi.org/10.5281/zenodo.23863986), a dated edition that freezes, for each of fifteen critical infrastructure sectors, a sector framework, a facility class and a reference event as they stood on 1 November 2026. The edition is not falsified by a later revision of this document, whose working master continues to move. It is a working draft, marked DRAFT on its cover.

1. Purpose, Scope and Position

1.1 Purpose

This instrument reads the collapse of the Francis Scott Key Bridge at Baltimore on 26 March 2024, when the containership Dali lost power, propulsion and steering and struck a pier of the central span, killing six members of a highway maintenance crew and closing the channel beneath. It reads the event back through DBA-TRAN-WY-FC2 (*Ways and Linehaul*), whose governing case it is (§1.4, §2.3).

1.2 Position and Scope

A reference event instrument, in the series convention, reads a real event backward through the design basis, adds no requirements, and returns findings for disposition. This one was issued under the Transportation Sector Framework v0.10, inherits the DBA-MA parent framework, and was conferred by the framework's §9.1, at v0.7. The scope runs from the Dali's departure shortly after midnight on 26 March 2024 to the restoration of the channel on 10 June 2024, with the National Transportation Safety Board's investigation to its board meeting of 18 November 2025 and the replacement, settlements and criminal proceedings to the record date of 8 October 2026. Times are Eastern Daylight Time.

On the author's decisions of 8 October 2026, no individual who was charged with an act, or whose act or omission bore on the event, is named: the Dali's crew, its pilots and the managers' staff are named by role, the six workers who died and the worker who survived are described and not named, and officials are named by role; organizations are named and every count is attributed to its publisher. No question of state attribution arises: on 26 March 2024 the Governor of Maryland and the Special Agent in Charge of the Federal Bureau of Investigation's Baltimore office stated that they had seen no evidence of a terrorist attack, and the Board determined a mechanical and electrical probable cause; this instrument states those as their authors' and makes none of its own. The Board's reports are primary (Section 2.1). The Board's probable cause is stated as the Board's determination; no finding of fault is made.

The span, the channel and the highway lie within the ways class (§1.3). The vessel is DBA-TRAN-CV-FC3's (*Conveyances and Fleet*), the port DBA-TRAN-ND-FC1's (*Nodes and Terminals*), and the Captain of the Port's closure and phased reopening of the waters DBA-TRAN-CntlCo's (*Traffic Management, Signaling and Control*). The Foundational Definitions are cited by designation, FD-REG holding their versions. Section references of the bare form §N are to *Ways and Linehaul* v0.1, the version this instrument read.

2. Sources and Attribution

2.1 Sources and Their Limit

The primary sources are the Board's reports MIR-25-40 and MIR-25-10, its investigation page and summary of owners' responses, the Corps of Engineers' announcement of 10 June 2024 and the owner's releases on the replacement; the numbered findings of MIR-25-40 and the docket are unread. The press is cited where used. No non-public material is used.

2.2 Where the Sources Differ

The predecessor sets out seven differences. On the bridge's vulnerability, the Authority said that "the collapse was the sole fault of the Dali and the gross negligence of her owners", against the Board's contributing cause (Section 3.6); each position is stated as its author's. The Attorney General of Maryland described the Board's report as finding the shipbuilder at fault; the Board does not determine fault. The ship's owner and manager settled the United States' and the State's claims, and no conclusion on liability is drawn from either. In May and June 2026 federal prosecutors charged the ship's managers, their technical superintendent and its chief engineer, as WYPR and Maryland Matters reported; the defendants are presumed innocent, the charges are stated as the prosecutors' allegations, and no finding is made on them.

3. The Event

3.1 The Bridge and the Channel

The bridge opened in 1977, carrying four lanes of Interstate 695 across the Patapsco River, with a vertical clearance of 185 feet (Maryland Transportation Authority). It was designed to the 1969 edition of the Association's specifications, which did not mention vessel collision, and was built with four dolphins and crushable fendering at Piers 17 and 18 (MIR-25-10). The Fort McHenry Federal Channel, 700 feet wide and 50 feet deep, runs under it. In 1980 the 390-foot Blue Nagoya struck Pier 17 and was stopped by the fendering; the Board puts its displacement at about one-tenth of the Dali's, and states that ocean-going vessels "have only grown larger since the Key Bridge's opening in 1977".

3.2 The Vessel and the Departure

The Dali, 984 feet long, displaced 112,383 metric tons (MIR-25-40).

3.3 The Loss of Power

On the Board's account in MIR-25-40, at 01:25:00, at 8.9 knots and about 3,200 feet from the bridge, a high-voltage breaker opened because a signal wire had come loose from a terminal block, where wire-label banding had prevented it from being fully inserted; the ship lost steering and its auxiliary pumps, and at 01:25:08 the main engine shut down. The breakers, in their

manual setting, were closed by hand at 01:25:58, after 58 seconds, where in the automatic setting power would have returned in about ten; the emergency generator connected at 01:26:10, after 70 seconds. At 01:27:04 the generators were disconnected for underperformance, which the Board ties to fuel pressure from a flushing pump, and the ship blacked out again until 01:27:36. At 01:29:09 the bow struck the northwest column of Pier 17 at 6.4 knots, about four minutes after the first blackout. The Board gives the pier's lateral capacity, from the Federal Highway Administration's calculation, as 5,509 kips, and estimates the impact at more than four times that; none of the four dolphins was struck (MIR-25-10). Pier 17 collapsed about thirteen seconds after the impact.

3.4 The Warning and the Clearance

At 01:25:59, a minute after the first blackout, the senior pilot telephoned the pilots' dispatcher and told the dispatcher to have local authorities shut down the bridge; at 01:26:44 the dispatcher called the Maryland Transportation Authority's duty officer; at 01:27:53 the duty officer told the officers at each end to close the bridge; and at 01:28:21 they blocked the lanes, the bridge being cleared of traffic about 48 seconds before the strike (MIR-25-40). The officer at the north end was holding the lanes until relieved. The Board states that the workers and the inspector on the bridge "were not notified of the Dali's emergency situation before the bridge collapsed", and that the bridge "was not equipped with a warning system to prevent motorists from driving onto the bridge in the event of a hazard".

3.5 The People on the Span

Seven workers of a contractor, with a subconsultant's inspector, were repairing potholes and were on break in their vehicles at the strike; six died (MIR-25-40).

3.6 The Board's Determination

The Board determined "that the probable cause of the contact of the containership Dali with the Francis Scott Key Bridge was a loss of electrical power (blackout), due to a loose signal wire connection to a terminal block stemming from the improper installation of wire-label banding, resulting in the vessel's loss of propulsion and steering close to the bridge. Contributing to the crew's inability to recover propulsion from the loss of electrical power was the limited time available due to the Dali's proximity to the bridge. Contributing to the collapse of the Francis Scott Key Bridge and the loss of life was the lack of countermeasures to reduce the bridge's vulnerability to collapse due to impact by ocean-going vessels, which could have been implemented if a vulnerability assessment had been conducted by the Maryland Transportation Authority as recommended by the American Association of State Highway and Transportation Officials. Also contributing to the loss of life was the lack of effective and immediate communications to notify the highway workers to evacuate the bridge" (MIR-25-40). It recommended motorist warning systems (H-25-28 to H-25-30) and emergency communications for bridge workers (H-25-31).

3.7 The Bridges the Board Named

In MIR-25-10 the Board found the bridge's annual frequency of collapse from vessel collision, by Method II of the Association's guide, to be 0.002921, almost thirty times the criterion of 0.0001 a year for a critical or essential bridge, and that no federal body can require an owner to make the assessment. By an operational screen it named 68 bridges, held by 30 owners, with no recent assessment; on its summary of 25 November 2025, 14 were reported above the criterion, 22 below and 32 still being calculated.

3.8 The Channel

The Captain of the Port closed the waters and, with the Unified Command, opened temporary channels in stages: the first on 2 April for shallow-draft vessels, one estimated by the Captain of the Port to carry about 15 per cent of the prior commercial activity, and, the day after the Dali was refloated on 20 May, a channel 50 feet deep. The channel was restored to its full dimensions on 10 June, seventy-six days after the collapse (MIR-25-40; Corps of Engineers). The Corps counts 1,587 responders from about 56 agencies.

3.9 The Port and the Highway

The port's flow was stranded while the channel was closed and recovered after. The highway's flow went through the harbor tunnels, which bar hazardous materials; that cargo had no harbor crossing, as WMAR reported.

3.10 The Replacement

The replacement is a cable-stayed bridge with a main span of 1,665 feet, a clearance of 230 feet and pier protection sized to the Association's current guidance (Maryland Transportation Authority). Its estimate rose from 1.7 to 1.9 billion dollars, made on the old bridge's design, to 4.3 to 5.2 billion dollars in November 2025. The largest procurement, for the main span, is to begin construction in the summer of 2027, and the owner has stated that a final schedule will not be known until the remaining contracts are awarded (Maryland Transportation Authority, 19 May and 25 August 2026).

3.11 Attribution and Later Events

The officials' statements of 26 March 2024 are at Section 1.2, as PolitiFact reported them; the proceedings of Section 2.2 are context.

4. Technical Decomposition

The event decomposes into initiating elements (I-1 to I-5) and effect elements (E-1 to E-6), as Table 4-1 sets out. Composed, they are the class's chain S in both ways the span carried (§3.2), realized through a vessel strike on a span over navigable water (§4.1), with the people on the

way harmed within it, and the channel's severance ended by repair and the highway's by replacement (§3.3).

Table 4-1. Event elements and their mapping. Section references are to Ways and Linehaul v0.1 unless another instrument is named.

Ref	Description	Provisions engaged	Finding
I-1	Loss of power and of propulsion and steering about 3,200 feet and four minutes from the span	§3.5, §4.1	F-4
I-2	Span designed to a 1969 specification, never assessed for vessel collision; annual frequency of collapse almost thirty times the criterion	§2.3, §3.5, §4.1, §8.4	F-1, F-10
I-3	Dolphins and fendering not in the vessel's path; impact more than four times the pier's lateral capacity	§4.1, §5.8	F-2
I-4	Vessels about ten times the displacement of the 1980 allision; 3,775 ocean-going transits in 2023	§4.1, §5.8	F-3
I-5	Warning through the pilots' dispatcher and the owner's duty officer; no motorist warning system; no means to reach the workers	§3.5, §4.1, §5.5, §7.2, §7.3 (CDB-4)	F-4, F-5, F-9
E-1	Pier 17 and the central spans collapsed; six workers killed; traffic stopped about 48 seconds before	§3.2, §3.5	F-4, F-5
E-2	Channel reopened in stages and fully restored after seventy-six days by a response drawn from many agencies and countries	§3.3, §4.6, §8.2	F-6
E-3	Highway severed; detour through tunnels that do not admit hazardous materials	§3.3, §8.1	F-8
E-4	Port tonnage 45.9 million tons in 2024 against 52.3 million in 2023	§3.3, §8.1; Nodes and Terminals §4.5	F-8
E-5	Replacement redesigned to current guidance; estimate up from 1.7–1.9 to 4.3–5.2 billion dollars; schedule not set	§2.3, §3.3, §4.2, §5.7	F-7, F-10
E-6	Sixty-eight bridges named; 14 above the criterion, 22 below, 32 not yet calculated by November 2025	§3.5, §4.1, §8.4	F-1

5. Tier Determination

The four characteristics are read as the class states them, against the Board's account.

5.1 Intentionality

Absent, on the Board's determination and the officials' statements; this instrument makes no finding beyond its sources.

5.2 Persistence

Absent. A single loss of power aboard one vessel.

5.3 Precision

Absent in the event, present in its effect.

5.4 Denial

Absent.

5.5 Determination

The event does not meet the threshold for the military-action extension on any of the four characteristics, on the Board's account and on the officials' statements of 26 March 2024. It is a non-adversarial loss of power aboard a vessel in confined water, the conveyance class's event, that realized at the span the class's postulated event of a vessel strike on a span over navigable water (Ways and Linehaul §4.1), in chain S for both of the ways the span carried. The provisional placement is confirmed. That is the tier determination this instrument returns for the DBA Reference Event Register.

5.6 The Constructed Analog

Kept in short form because F-2 and F-9 rest on it. The class's third scenario is the strategic span as the target (§5.5). It would need less than the scenario imagines: a laden containership without propulsion at 6.4 knots brought down a pier whose capacity was less than a quarter of the force it met, and a ship under control could choose the pier, the angle and the speed, passing between protections placed for an aberrant ship's likely path. It would meet a warning made of people, every second of which was bought by a call from the ship itself; a ship whose master or pilot intended the strike would make no call. And it would meet a replacement no plan had prepared. It is not a claim that the event was such an act.

6. Characterization Against the Design Basis

6.1 §1.4, §2.3 and §10.7 The Class's Account of the Event

Held, with a gap. Accurate in direction; the probable cause in full, the Board's screen, the replacement's figures, the channel's description and the guide's title need qualifying. Finding F-10, now an erratum.

6.2 §1.3, §2.1 and §3.2 The Span That Severed Two Ways

Held. One pier's failure severed the highway and the channel and killed the six workers.

6.3 §3.5 and §4.1 The Vessel-Collision Risk of a Span

Held, with a gap. The parameter and criterion are confirmed; the class does not say how the spans to which they apply are found, and the Board applied an operational screen. Finding F-1.

6.4 §4.1 and §5.8 Pier Protection as Built

Held, with a gap. Protection rated in place was not struck; the class credits it by presence, not by what it intercepts. Finding F-2.

6.5 §4.1 and §5.8 The Vessels That Grew

Held, with a gap. The assessment is renewed when the channel changes; the vessels grew through a channel that had not. Finding F-3.

6.6 §3.5, §4.1 and §7.2 The Warning-to-Clearance Time

Held, with a gap. The time was held for the motorists by 48 seconds and not at all for the workers. Finding F-4.

6.7 §3.2, §3.5 and §7.3 The People on the Span

Held, with a gap. The workers were a contractor's, on break in their vehicles, and the officer who could have reached them was holding the lane. Finding F-5.

6.8 §3.3, §4.6 and §8.2 Severance Time: The Channel's Repair

Held, with a gap. The channel reopened in stages, each with a share of flow. Finding F-6.

6.9 §3.3, §4.1, §4.2 and §5.7 Severance Time: The Span's Replacement

Held, with a gap. The measure is confirmed; the bridge being built is not the bridge that fell, and the first estimate after a collapse is made before that bridge has been designed. Finding F-7.

6.10 §3.3 and §8.1 Corridor Stranded Flow

Held, with a gap. Hazardous materials, which the tunnels do not admit, had no crossing. Finding F-8.

6.11 §5.1, §5.5 and §5.6 The Military-Action Extension

Held, with a gap. The scenario is confirmed; the warning came from the vessel itself, which the deliberate case cannot count on. Finding F-9.

6.12 §8.3 and §8.4 Boundaries and the Regulatory Context

Held. The event divides at the strike as the class draws the line, and no regime required the assessment (§8.4). Part of F-1.

6.13 Framework §4.2 The Sector Principles

Held, with the gaps F-2 to F-6 and F-8 state.

6.14 §6.3 The AI-Governance Design Basis

Not exercised. No component within the scope of *Layer Decomposition and Determinism* §4.3 bore on the event.

6.15 Summary of Design Basis Performance

Of the fourteen readings, eleven Held, with a gap, the class's account among them; two Held; and the AI-governance provisions were Not exercised.

7. Findings

Section references are to *Ways and Linehaul* v0.1 unless another instrument is named.

7.1 F-1 — The Spans at Risk Must First Be Found

The Key Bridge's annual frequency of collapse was almost thirty times the criterion; its owner had not made the assessment, no federal authority could require it, and neither its federal permit nor its inspections asked the question, the Coast Guard stating in 2026 that it has no statutory authority over the structural integrity of bridges. Of the 68 bridges the Board named, 14 were reported above the criterion by November 2025 (Section 3.7). The finding confirms the vessel-collision parameter, its criterion and the class's decision to make the assessment a requirement (§3.5, §4.1, §8.4), and is that the class should state, as the requirement's first step, a screen of the kind the Board used (waterways used by ocean-going vessels, piers in the water, clearance and transits above stated values), with the inventory held by the owner and the Federal Highway Administration and no bridge leaving the requirement on the screen alone; cite the Board's review and the owners' responses as the population's first measure; and name, in the framework's category for the severance of a critical span, the assessment of spans built before the guidance (framework §4.3).

Affected: §3.5, §4.1, §8.4; framework §4.3. Disposition: R-1.

7.2 F-2 — Pier Protection Is Credited by What It Intercepts

The bridge had four dolphins and crushable fendering at its channel piers, which had stopped a 390-foot ship in 1980; the Dali struck Pier 17 without touching a dolphin, at a force the Board estimates at more than four times the pier's lateral capacity, and the Board found that the

dolphins' locations and size did not fully protect the piers, two years after an inspection had rated the protection in place and functioning properly. The finding confirms the required function of protecting or strengthening the piers (§4.1) and is that the class should state that pier protection is credited in the assessment only for what it is shown, as built and for the vessels that use the channel now, to intercept and stop on the paths an aberrant vessel takes, and that protection present but not shown to intercept is not credited; and that §5.8 should list among its relaxations the crediting of pier protection by its presence. For a span whose determination is positive, the same showing is made for a vessel under control that chooses its pier (§5.5).

Affected: §4.1, §5.5, §5.8. Disposition: R-2.

7.3 F-3 — The Assessment Is Renewed When the Vessels Change

The ship that struck Pier 17 in 2024 was about ten times the displacement of the ship that struck it in 1980, and the Board attributes the growth to advances in shipping, among them the expansion of the Panama Canal in 2016, through a channel deepened to 50 feet after the bridge opened, and not changed in its dimensions in the years in which, on the Board's account, the largest growth came. The class renews the assessment when a channel is deepened or widened (§4.1). The finding confirms that the assessment is made for the vessels that use the channel now (§4.1) and is that the class should add, as triggers for renewal, a change in the size, displacement or number of the vessels using the channel, measured from the data of the Automatic Identification System, and a stated interval in any case; and that §5.8 should name an assessment not renewed since the vessels changed beside the assessment made for the ships of the year of design.

Affected: §4.1, §5.8. Disposition: R-3.

7.4 F-4 — The Warning Time Begins at the Loss of Control, and Its Path Is Named

The ship lost power at 01:25:00 and struck at 01:29:09; the pilot's call left at 01:25:59, and traffic was stopped at 01:28:21, after two relays, 48 seconds before the strike (Section 3.4); the pilots' protocol named the Coast Guard first, the dispatcher called the owner first, and the bridge had no motorist warning system. The finding confirms the warning-to-clearance time as the parameter that decided who lived (§3.5) and the class's reading (§7.2), and is that the class should state the time from the loss of control aboard, not from the first warning; name every link of the path by which a warning reaches the means of closing a span, the pilots' dispatch and any vessel traffic service included, with the first call fixed in each party's protocol; make the means of closure operable on the first warning by the first party to receive it, as the Board's recommendations on motorist warning systems would; and read "the word of the vessel" in §4.1 as the word of the vessel, its pilot or the pilots' dispatch, or the vessel traffic service.

Affected: §3.5, §4.1, §7.2; noted to Traffic Management, Signaling and Control and Conveyances and Fleet. Disposition: R-4.

7.5 F-5 — The Workers on the Span Are Reached Apart from the Closure

Seven workers of a contractor, with a subconsultant's inspector, were on break in their vehicles; the officer at the north end was holding the lanes until relieved, and no means reached them. The Board found that the lack of effective and immediate communications to notify them contributed to the loss of life, and recommended that the work-zone standard require such communications on bridges over navigable waters. The finding confirms the class's inclusion of workers (§3.2, §3.5) and its reading of CDB-4 (§7.3), and is that the class should state that the requirement reaches every person the owner admits to a critical span, contractors and inspectors included; that the owner's permit for work on such a span carries the means of warning as a condition; and that the clearance of the people on the span is a step separate from the closure of the lanes, staffed so that neither waits on the other, and exercised and measured under CDB-7.

Affected: §3.2, §3.5, §7.3 (CDB-4, CDB-7). Disposition: R-5.

7.6 F-6 — The Channel's Repair Is Counted in Stages, and Its Capacity Is Stated

The channel reopened in stages, the first for shallow draft on 2 April, one of 20 feet expected to carry about 15 per cent of the commercial activity, one of 50 feet the day after the ship was refloated, and the full channel on 10 June, with a response drawn from about 56 agencies and from around the world (Section 3.8). The finding confirms repair time as the measure for the channel (§3.3) and the counting of a partial channel at the ship it admits (§4.6), and is that the class should state the repair time of a channel of consequence as a sequence of stages, each with the largest ship it admits and its share of the design flow; that where the obstruction is a fallen span on a vessel, the stages are set by the removal of the span from the vessel and of the vessel from the channel; and that the salvage, heavy-lift and wreck-removal capacity on which the repair time rests, which §8.2 says no instrument names, is stated by the channel's owner as a quantity with its source and its time to arrive (Transportation Sector Framework §4.2).

Affected: §3.3, §4.6, §8.2. Disposition: R-6.

7.7 F-7 — Replacement Time Is the Time to Build the Bridge Current Standards Require

The replacement has a longer main span, a higher clearance and a pier protection system sized to current guidance; the owner attributes the growth of its estimate from 1.7–1.9 to 4.3–5.2 billion dollars to the cost of materials and to that protection, which the preliminary estimate did not include, and the Authority's executive director said the first estimate had been made without design work and on the old bridge's design. Since May 2026 the owner has given no schedule. The finding confirms replacement time as the measure that sizes a critical span's protection, the recording of cost growth beside it (§3.3), and the residual of §5.7, and is that the class should state that the replacement time of a critical span is estimated, before it is needed, for the

replacement that current standards would require, with its protection, and not for the structure that stands; that the critical-span inventory of §3.5 records that estimate and its basis; that the design-basis duration of §4.1 and the residual of §5.7 are stated in those terms; and that the class should cite the owner's estimates with the record date.

Affected: §3.3, §3.5, §4.1, §4.2, §5.7. Disposition: R-7.

7.8 F-8 — An Alternative Absorbs Only What It Admits

The highway's traffic crossed the harbor through the two tunnels, which do not admit hazardous materials; the bridge had been the only harbor crossing for that cargo, which went around the western side of the Beltway or through the city, on the Maryland Motor Truck Association's account. The port's tonnage fell from 52.3 million tons in 2023 to 45.9 million in 2024, the node's count. The finding confirms the class's computation of corridor stranded flow and its counting of the channel's nodes by the nodes (§3.3; Nodes and Terminals §4.5) and is that the class should state that an alternative is credited with the flow it admits by cargo class, height, weight and hazard; that a way which is the sole path for one class of cargo is counted as the sole path for that class, and the sole-path elevation of §8.1 applied by class; and that §5.8 should list the crediting of an alternative with flow it does not admit.

Affected: §3.3, §5.8, §8.1. Disposition: R-8.

7.9 F-9 — In the Deliberate Case No One Calls

Every second of the clearance at Baltimore was bought by a call made from the ship within a minute of its loss of power. A vessel whose master or pilot intended the strike would make no call, and could choose the pier, the angle and a speed above 6.4 knots, at which a laden containership brought down a pier whose capacity was less than a quarter of the force it met. The finding confirms the scenario of the strategic span as the target (§5.5) and the Sustained Corridor Condition's requirement that spans be clear of people when threatened (§5.6), and is that the class should state that, for a span whose determination is positive, the warning-to-clearance time is measured from the detection of the approach by the way's own means or by the control class, not from a warning given by the vessel; that the means of closure act on that detection; and that the protection of §5.5 is shown for a vessel under control that chooses its pier (F-2). It carries, at v0.2, the attacks on the Kerch Strait bridge (DBA-TRAN-RE-11), recast on the author's ruling: in October 2022 one act damaged both the road and the rail structures standing side by side, so that the critical-span inventory should record whether a span's modes and carriageways stand on separate structures, credit the partial service they give, and not credit them as independent of a deliberate act at one place (framework §4.2); and the bridge was struck again ten weeks after its second rail track reopened, so that the adequacy test of §5.3 should count a second strike on the span during or soon after restoration, with the alternative's capacity stated as a measured quantity (Kerch Strait Bridge F-2, F-4).

Affected: §1.3, §3.5, §5.2, §5.3, §5.5, §5.6; framework §4.2. Disposition: R-9.

7.10 F-10 — Withdrawn at v0.2

Its substance, qualifying the class's account of the event at §1.4, §2.3, §3.5, §8.4 and §10.7, goes to *Ways and Linehaul* as Erratum 1 of Project 11's note to Project 9 of 10 October 2026 (batch 19).

8. Recommendation Register

Priority is on the series' one-to-three scale, 1 the most consequential. R-1 to R-9 dispose to *Ways and Linehaul*, the framework element of R-1 also to the Transportation Sector Framework; R-4 is noted to *Traffic Management, Signaling and Control* and *Conveyances and Fleet*. R-9 now carries the Kerch Strait bridge's R-2 and R-4. R-10 is withdrawn, its substance going to the class as an erratum.

ID	Recommendation	Affected	Direction	Pri
R-1	State the screen that finds the spans at risk	§3.5, §4.1, §8.4; framework §4.3	Add an operational screen of the Board's kind as the requirement's first step, the inventory held by the owner and the Federal Highway Administration, no bridge leaving the requirement on the screen alone; name the assessment of pre-guidance spans in the framework.	1
R-2	Credit pier protection by what it intercepts	§4.1, §5.5, §5.8	Credit protection only for what it is shown, as built and for the current vessels, to intercept and stop on an aberrant vessel's paths; list crediting by presence among the relaxations; for a positive determination, show it for a vessel that chooses its pier.	1
R-3	Renew the assessment when the vessels change	§4.1, §5.8	Add a change in the size, displacement or number of the vessels using the channel, measured from Automatic Identification System data, and a stated interval as triggers for renewal; list an assessment not renewed since the vessels changed among the relaxations.	2
R-4	Measure the warning from the loss of control, and name its path	§3.5, §4.1, §7.2; noted to Traffic Management, Signaling and Control and Conveyances and Fleet	Run the time from the loss of control; name every link of the warning path with the first call fixed; make closure operable on the first warning; read "word of the vessel" to include the pilots.	1
R-5	Reach the workers apart from the closure	§3.2, §3.5, §7.3 (CDB-4, CDB-7)	Extend the requirement to every person admitted to a critical span; make the means of warning a permit condition; make the clearance of people a step separate from the closure of the lanes.	1
R-6	Count the channel's repair in stages, and	§3.3, §4.6, §8.2	State repair time as stages, each with the largest ship admitted and its share of flow; state the salvage and	2

	state its capacity		heavy-lift capacity with its source and time to arrive.	
R-7	Estimate replacement for the bridge current standards require	§3.3, §3.5, §4.1, §4.2, §5.7	Estimate replacement time for the bridge current standards require, with its protection, recorded in the inventory; state §4.1's duration and §5.7's residual in those terms.	2
R-8	Credit an alternative with what it admits	§3.3, §5.8, §8.1	Credit an alternative by cargo class, height, weight and hazard; count a way that is the sole path for one class as such and apply the sole-path elevation by class; list crediting an alternative with flow it does not admit among the relaxations.	2
R-9	Measure the deliberate case from detection; count adjacency and the second strike	§1.3, §3.5, §5.2, §5.3, §5.5, §5.6; framework §4.2	For a positive determination, measure the warning-to-clearance time from detection of the approach by the way's own means or the control class, with the means of closure acting on that detection. Carries DBA-TRAN-RE-11 R-2 and R-4 at v0.2: record separate structures in the critical-span inventory, credit their partial service and not their independence of a deliberate act at one place; count a second strike during or after restoration in the adequacy test, with the alternative's capacity measured.	1
R-10	Withdrawn at v0.2	§1.4, §2.3, §3.5, §8.4, §10.7	The qualification of the class's account goes to Ways and Linehaul as Erratum 1 of Project 11's note of 10 October 2026 (batch 19), and is withdrawn with F-10.	—

9. Open Items

OI-1, the register row, is closed: Project 0 entered the tier determination of Section 5.5. OI-2 stands: the replacement's schedule, the named bridges' assessments, the claims and proceedings are open, bearing on F-1, F-4, F-5 and F-7. OI-3 stands: the class has not taken in this instrument, and conforms at its next revision, with the erratum, stating its events as conferred.

10. Standing of This Document

10.1 What This Instrument Establishes

It modifies neither class nor framework, and makes no finding on cause beyond the Board's determination, none of fault, and none of intent.

10.2 Relationship to the Class's Other Events

This is the ways class's governing case. The attacks on the Kerch Strait bridge, recast on the author's ruling, are the military-action case, whose adjacency and second-strike findings F-9 now carries.

10.3 Relationship to the Series' Other Instruments

The ship's part, the loose wire, the breakers left in their manual setting, the flushing pump that did not restart by itself, the slow emergency generator and the four minutes the ship had, is the case of *Conveyances and Fleet* §4.1, whose control recovery margin is spent from the same four minutes as the way's warning-to-clearance time. This instrument notes to that class that the timings its §3.3 takes from the Board's board summary, a first blackout of fifty-eight seconds where the automatic setting would have restored power in about ten, and an emergency generator that connected after seventy seconds, are confirmed by MIR-25-40.

References

The sources named in Sections 1.2 to 2.2, the predecessor keeping the full list; the DBA instruments named in the text, with DBA-MA-CDB, the Foundational Definitions and FD-REG, and the DBA-MA Parent Framework.

Revision History

Version 0.1, October 2026: issued in Project 9 under the Transportation Sector Framework v0.10 on the author's decisions of 8 October 2026, with the tier determination of Section 5.5; F-1 to F-10, R-1 to R-10.

Version 0.2, 10 October 2026: migrated to Project 11 under the determination of 9 October 2026 (*Reference Events Move to Their Own Project*), compressed under the Reference Event Compression Standard v0.3 and brought to house style. Cut: the general recitals, to Section 1.2; sourcing and the seven contested points, to what is primary and their resolution, the charges kept as allegations; the replacement-cost narrative and tonnage figures; the analog, to a short form; Section 6, to the standard's four verdicts. Section 2 is renumbered. Sections 3, 5, 6 and 10 keep their numbering (DBA-TRAN-UC-2 cites §3.3, §3.4, §3.10 and §6.9, and DBA-TRAN-UC-3 cites §10.3). Received: on the author's ruling recasting the attacks on the Kerch Strait bridge (DBA-TRAN-RE-11), its F-2 and F-4 are carried by F-9 in an added sentence, and its R-2 and R-4 by R-9; the sending end is Project 9's. Stub: F-10, an erratum to *Ways and Linehaul* (batch 19 note), with R-10. F-1, F-4 and F-5, which no instrument cites by number, are compressed with every ask kept. Unchanged: the designator, title, tier determination, attribution statements, sources, F-2, F-3, F-6 to F-8 and R-1 to R-8. OI-1 closed. Verified: every finding and recommendation number present at v0.1 is present here as an entry or a stub.